

Aircraft Fuel Tank & Inerting System Simulation

Study Notes: Definitions, Concepts, Formulae, Code Mapping, Assumptions and Simulation Flow

These notes are based on the project code and explanations provided in this conversation. The model is a reduced-order educational simulation, not an Airbus-certified or aircraft-specific fuel/inerting model.

1. What the project is modelling

The simulation couples four main subsystems:

- **Atmosphere / flight profile:** determines altitude, flight phase, ambient temperature and ambient pressure.
- **Fuel tank:** tracks fuel mass, fuel volume, ullage volume and ullage temperature.
- **Inerting system:** tracks ullage O_2 and N_2 moles and models nitrogen-enriched air (NEA) injection, ambient-air entry and venting.
- **PI controller:** measures ullage O_2 concentration and changes NEA flow to move O_2 toward a target.

The simulation repeats these calculations at every timestep and stores the state in a Pandas DataFrame.

2. Key terminology and definitions

Term	Definition
Fuel mass	Mass of liquid fuel remaining in the tank, in kg.
Fuel density	Fuel mass per unit volume, kg/m^3 .
Fuel volume	Volume occupied by the liquid fuel.
Ullage	The gas-filled space above the liquid fuel in the tank.
Ullage volume	Tank volume not occupied by liquid fuel.
Ullage gas	Gas mixture occupying the ullage; this simplified model tracks O_2 and N_2 only.
O_2 fraction	Mole fraction of oxygen in the ullage: O_2 moles divided by total O_2 + N_2 moles.
NEA	Nitrogen-enriched air injected into the ullage to dilute oxygen.
Inerting	Reducing/controlling ullage oxygen concentration by introducing an oxygen-depleted gas.
Ambient air	External air assumed here to contain 20.95% O_2 and 79.05% N_2 .
Mole / mol	Amount of substance used by the ideal-gas model.
Ideal gas law	Relationship between pressure, volume, temperature and amount of gas.
PI controller	Feedback controller using proportional and integral terms to calculate a control input.
Setpoint / target	Desired O_2 fraction; the default model uses 0.12 = 12%.
Saturation	Limiting a controller output to a physical minimum/maximum.
Anti-windup	Preventing the integral term from accumulating while the actuator is saturated.
ISA	International Standard Atmosphere model used here for simplified pressure/temperature calculations.

3. Constants and physical quantities

Atmosphere constants used in the code:

- $R = 8.314 \text{ J/(mol}\cdot\text{K)}$: universal gas constant.
- $T_{SL} = 288.15 \text{ K}$: sea-level standard temperature.
- $P_{SL} = 101325 \text{ Pa}$: sea-level standard pressure.
- $L = 0.0065 \text{ K/m}$: temperature lapse-rate magnitude used by the model.
- $g_0 = 9.80665 \text{ m/s}^2$: standard gravitational acceleration.
- $M_{air} = 0.0289644 \text{ kg/mol}$: molar mass of air.

Tank defaults:

- Tank volume = 18 m³.
- Initial fuel = 12,000 kg.
- Fuel density = 800 kg/m³.
- Initial ullage temperature = 288.15 K.
- Thermal time constant = 900 s.
- Burn rates: climb 1.1 kg/s, cruise 0.6 kg/s, descent 0.3 kg/s, ground 0 kg/s.

Inerting defaults:

- NEA O₂ fraction = 0.05 (5%).
- Maximum sea-level inert flow = 0.15 mol/s in the model.
- Minimum flow fraction = 0.2 of the sea-level maximum.

Controller defaults:

- Target O₂ = 0.12.
- K_p = 2.0.
- K_i = 0.0008.
- Minimum control output = 0.

4. Flight profile

4.1 Flight phases

- Climb: from t = 0 to 1200 s.
- Cruise: from 1200 s to 9600 s.
- Descent: from 9600 s to 10800 s.
- Ground: after the total flight time.

$$\begin{aligned}\text{Total flight time} &= t_{\text{climb}} + t_{\text{cruise}} + t_{\text{descent}} \\ &= 1200 + 8400 + 1200 = 10800 \text{ s} = 180 \text{ min}\end{aligned}$$

Default cruise altitude is 10,668 m, approximately 35,000 ft.

4.2 Altitude model

During climb, altitude increases linearly:

$$h(t) = h_{\text{cruise}} \times (t / t_{\text{climb}})$$

During cruise:

$$h(t) = h_{\text{cruise}}$$

During descent:

$$\begin{aligned}\text{fraction} &= (t - t_{\text{cruise_end}}) / (t_{\text{total}} - t_{\text{cruise_end}}) \\ h(t) &= h_{\text{cruise}} \times (1 - \text{fraction})\end{aligned}$$

This is a simplified flight profile rather than a real aircraft trajectory.

5. ISA atmosphere model

5.1 Temperature

For the model's tropospheric region, altitude is first limited to 11,000 m.

$$h = \min(h_m, 11000)$$

$$T(h) = T_{11} - Lh$$

Meaning: temperature decreases linearly with altitude in this simplified model. Above 11 km, the code holds the temperature at the 11 km value rather than extrapolating the same linear equation.

5.2 Pressure

$$P(h) = P_{11} \times (T/T_{11})^{[g M_{\text{air}} / (R L)]}$$

Pressure decreases with altitude. In the simulation, this pressure is also used as the simplified tank pressure.

5.3 Important distinction

The ISA equations provide an atmospheric environment for the simulation. They are not a detailed aircraft environmental-control or fuel-tank pressure model.

6. Fuel tank model

6.1 Fuel volume

$$V_{\text{fuel}} = m_{\text{fuel}} / \rho_{\text{fuel}}$$

where m_{fuel} is fuel mass and ρ_{fuel} is fuel density.

6.2 Ullage volume

$$V_{\text{ullage}} = \max(V_{\text{tank}} - V_{\text{fuel}}, 10^{-3})$$

The max operation prevents the ullage volume from reaching zero in the numerical model.

6.3 Fuel burn

$$m_{\text{fuel,new}} = \max(m_{\text{fuel,old}} - \text{burn_rate} \times dt, 0)$$

Fuel mass cannot become negative.

6.4 Thermal response

$$T_{\text{ullage,new}} = T_{\text{ullage,old}} + [(T_{\text{ambient}} - T_{\text{ullage,old}})/\tau] \times dt$$

τ is the thermal time constant. A larger τ means slower temperature response. The ullage temperature therefore approaches ambient temperature gradually rather than instantly.

7. Ullage and ideal gas law

The ullage is the gas space created as fuel does not completely fill the tank. As fuel burns, fuel volume decreases and ullage volume generally increases.

7.1 Ideal gas law

$$PV = nRT$$

Definitions:

- P = gas pressure, Pa.
- V = gas volume, m^3 .
- n = amount of gas, mol.
- R = universal gas constant, $\text{J}/(\text{mol}\cdot\text{K})$.
- T = absolute temperature, K.

7.2 Rearranged for total gas moles

$$n = PV / (RT)$$

This equation is used to determine the total number of moles required for the current ullage state.

8. O₂ and N₂ composition

The simplified model represents the ullage using two gas species: oxygen and nitrogen.

8.1 Initial composition

At initialization, the model assumes ambient-air composition:

```
x_O2,initial = 0.2095
x_N2,initial = 1 - 0.2095 = 0.7905
n_O2 = n_total × x_O2
n_N2 = n_total × (1 - x_O2)
```

8.2 Oxygen fraction

```
x_O2 = n_O2 / (n_O2 + n_N2)
```

Example: $x_{O_2} = 0.12$ means 12% of the tracked ullage gas is oxygen by mole fraction.

9. Inerting / NEA model

9.1 What NEA does

Nitrogen-enriched air has a lower O₂ fraction than ambient air. Adding it to the ullage dilutes oxygen concentration.

In this model:

```
x_O2,NEA = 0.05
```

Therefore, each mole of NEA contributes:

```
Δn_O2 = n_NEA × 0.05
Δn_N2 = n_NEA × 0.95
```

9.2 Flow to moles

```
n_NEA = inert_flow × dt
```

If `inert_flow` is in mol/s and `dt` is in seconds, the result is moles added during the timestep.

9.3 Altitude-dependent flow limit

```
ratio = max(P_ambient/P_ref, minimum_flow_fraction)
max_flow = max_flow_sea_level × ratio
```

The code uses ambient pressure as a simplified proxy for reduced bleed/inert-gas supply capability with altitude.

9.4 External gas exchange: f_{ext}

```
f_ext = (n_target - n_previous) - n_NEA
```

This is not extra NEA. It is the remaining gas exchange needed after NEA injection has already been accounted for.

- $f_{ext} > 0$: additional ambient air enters.
- $f_{ext} < 0$: gas leaves/vents.
- $f_{ext} = 0$: NEA exactly supplies the required total-gas change.

When ambient air enters, the model assigns it 20.95% O₂ and 79.05% N₂. When gas leaves, it leaves at the current well-mixed ullage composition.

9.5 Gas-law target

```
n_target = P_ullage × V_ullage / (R × T_ullage)
```

This is the total gas amount required by the ideal gas law for the current pressure, volume and temperature.

9.6 Numerical rescaling

```

scale = n_target / n_total_new
n_O2,new = n_O2,new × scale
n_N2,new = n_N2,new × scale

```

This forces the final total gas amount to satisfy $PV = nRT$. Because both species are multiplied by the same factor, the O_2 fraction is not changed by the rescaling itself.

10. PI controller

10.1 Purpose

The controller creates a feedback loop: measure O_2 → compare with target → calculate error → command NEA flow → change O_2 → measure again.

10.2 Error

$$e(t) = x_{O_2, \text{measured}} - x_{O_2, \text{target}}$$

The sign is deliberate. If O_2 is above target, e is positive and the controller increases inert-gas flow.

10.3 PI equation

$$u = K_p \times e + K_i \times \int e \, dt$$

In the discrete simulation, the integral is accumulated as:

$$I_{\text{new}} = I_{\text{old}} + e \times dt$$

So the controller has two parts:

- **Proportional (P):** reacts to the current O_2 error.
- **Integral (I):** accumulates persistent error and helps remove steady-state offset.

10.4 Discrete implementation

```

u_unsat = Kp × error + Ki × integral
u = clamp(u_unsat, u_min, u_max)

```

The actual maximum can change with altitude because the controller receives the current inerting-system flow limit.

10.5 Saturation

A physical actuator cannot provide unlimited flow. The requested controller output is therefore constrained between minimum and maximum values.

10.6 Anti-windup

The model updates the integral only when the controller is not saturated:

```

if u == u_unsat: integral += error × dt

```

If the actuator is already at a limit, the integral is frozen. This prevents the integral term from growing excessively while the requested command cannot be physically delivered.

11. Main simulation loop

The simulation performs the following sequence every timestep:

- Calculate current time.
- Get altitude from the flight profile.
- Get flight phase.
- Calculate ambient temperature from ISA.
- Calculate ambient pressure from ISA.

- Measure current ullage O_2 fraction.
- Calculate PI-controller NEA flow command.
- Update fuel mass and ullage temperature.
- Recalculate ullage volume.
- Update O_2/N_2 moles using the inerting model.
- Store all relevant variables in a record.
- Repeat until the flight ends.

11.1 Data dependency flow

time → altitude/phase → atmosphere → tank state → O_2 measurement → PI controller → inerting → next state

The controller therefore does not directly change O_2 . It changes NEA flow; the inerting/gas model then determines how that flow changes the gas composition.

12. DataFrame outputs

Each timestep is stored with the following variables:

Column	Meaning	Typical unit
t_s	Simulation time	s
t_min	Simulation time	min
phase	Climb/cruise/descent/ground	—
altitude_m	Aircraft altitude	m
fuel_mass_kg	Fuel remaining	kg
ullage_volume_m3	Gas-space volume	m ³
tank_pressure_Pa	Simplified tank pressure = ambient	Pa
tank_temperature_K	Ullage temperature	K
o2_fraction	Ullage O_2 mole fraction	fraction
inert_flow_mol_s	Actual NEA flow	mol/s
fuel_burn_kg_s	Fuel burn rate	kg/s

13. Code-to-concept map

Code	Concept
isa_temperature(h)	Atmospheric temperature model
isa_pressure(h)	Atmospheric pressure model
FlightProfile.altitude(t)	Flight trajectory approximation
FlightProfile.phase(t)	Flight-phase logic
fuel_volume()	Mass-to-volume conversion
ullage_volume()	Tank geometry / remaining gas space
FuelTank.step()	Fuel depletion + thermal lag
$P V = n R T$	Ideal gas law
o2_fraction()	Gas composition measurement
max_flow()	Altitude-dependent actuator/supply limit
inert_moles = flow × dt	Continuous flow converted to timestep amount
f_ext	Ambient inflow / venting required by gas-law closure
PI controller	Closed-loop feedback control
saturation	Physical actuator limitation

Code	Concept
anti-windup	Integral protection during saturation
DataFrame	Time-history storage and analysis

14. Worked numerical examples

Example A: initial ullage

$\text{Fuel volume} = 12000 / 800 = 15 \text{ m}^3$
 $\text{Ullage volume} = 18 - 15 = 3 \text{ m}^3$
 $n_{\text{total}} = (101325 \times 3) / (8.314 \times 288.15) \approx 127 \text{ mol}$

The exact value is approximately 127 mol for the model's initial conditions.

Example B: O₂ and N₂ split

$n_{\text{O}_2} = 127 \times 0.2095 \approx 26.6 \text{ mol}$
 $n_{\text{N}_2} = 127 \times 0.7905 \approx 100.4 \text{ mol}$
 $x_{\text{O}_2} = 26.6 / (26.6 + 100.4) \approx 0.2095$

Example C: controller error

$\text{Measured } O_2 = 0.18$
 $\text{Target } O_2 = 0.12$
 $\text{error} = 0.18 - 0.12 = +0.06$

Positive error means the ullage is above the target O₂ level, so the controller attempts to increase NEA flow.

Example D: NEA addition

If flow = 0.10 mol/s and dt = 1 s:
 $n_{\text{NEA}} = 0.10 \times 1 = 0.10 \text{ mol}$
 At 5% O₂: $\Delta n_{\text{O}_2} = 0.005 \text{ mol}$
 $\Delta n_{\text{N}_2} = 0.095 \text{ mol}$

15. Physical interpretation of the main plots

- **Fuel mass vs time:** should generally decrease; slope changes according to climb/cruise/descent burn rates.
- **Ullage volume vs time:** should generally increase as fuel is consumed.
- **Tank pressure vs time:** follows ambient pressure in this simplified vented-tank assumption, so it decreases during climb and rises during descent.
- **Tank temperature vs time:** follows ambient temperature with a thermal lag.
- **O₂ fraction vs time:** should respond to NEA injection, gas exchange and changing ullage conditions; the controller attempts to move it toward the target.
- **Inert flow vs time:** reflects controller demand and the altitude-dependent maximum available flow.

16. Model assumptions and limitations

- Tank geometry is represented only by a total volume; detailed tank geometry is not modelled.
- Tank pressure is assumed equal to ambient pressure.
- Only O₂ and N₂ are tracked; water vapour, fuel vapour and other species are ignored.
- NEA composition is fixed at the chosen model value.

- The maximum inert flow is represented by a simplified pressure-based derating, not a detailed bleed-air/membrane model.
- Ambient air is fixed at 20.95% O₂ and 79.05% N₂.
- External gas exchange is inferred to satisfy the ideal-gas constraint.
- The 11 km ISA temperature clamp is a model boundary.
- Fuel density is treated as constant.
- Burn rates are fixed by flight phase.
- The model is reduced-order and intended for simulation/learning, not certification or direct aircraft design decisions.

17. What the model demonstrates

- Coupling between flight conditions and fuel-tank state.
- Mass conservation at a reduced-order gas-mixture level.
- Use of the ideal gas law in a dynamic simulation.
- Gas-mixture composition tracking.
- Feedback control of a physical quantity.
- Actuator saturation and anti-windup.
- Time-stepping and state updates.
- Data logging and post-processing with Pandas.

18. Interview-ready explanation

30-second version: “I built a reduced-order Python simulation of an aircraft fuel-tank inerting system. The model generates a simplified climb-cruise-descent profile, uses an ISA atmosphere to obtain pressure and temperature, models fuel burn and ullage growth, and tracks O₂/N₂ in the ullage using the ideal gas law. A PI controller measures ullage O₂ concentration and commands nitrogen-enriched-air flow to move the concentration toward a target. I also included an altitude-dependent flow limit and anti-windup, then logged the time history for analysis and plotting.”

19. Formula cheat sheet

Purpose	Formula
Fuel volume	$V_{\text{fuel}} = m_{\text{fuel}} / \rho_{\text{fuel}}$
Ullage volume	$V_{\text{ullage}} = V_{\text{tank}} - V_{\text{fuel}}$
Ideal gas law	$PV = nRT$
Gas moles	$n = PV / RT$
O ₂ fraction	$x_{\text{O}_2} = n_{\text{O}_2} / (n_{\text{O}_2} + n_{\text{N}_2})$
O ₂ moles	$n_{\text{O}_2} = n_{\text{total}} \times x_{\text{O}_2}$
N ₂ moles	$n_{\text{N}_2} = n_{\text{total}} \times (1 - x_{\text{O}_2})$
NEA moles per timestep	$n_{\text{NEA}} = \text{flow} \times dt$
External exchange	$f_{\text{ext}} = (n_{\text{target}} - n_{\text{previous}}) - n_{\text{NEA}}$
ISA temperature	$T = T_{\text{sea}} - Lh$
ISA pressure	$P = P_{\text{sea}} (T/T_{\text{sea}})^{gM/(RL)}$
PI error	$e = x_{\text{O}_2, \text{measured}} - x_{\text{O}_2, \text{target}}$
PI control	$u = K_p e + K_i \int e \, dt$
Discrete integral	$I_{\text{new}} = I_{\text{old}} + e \, dt$
Thermal update	$T_{\text{new}} = T_{\text{old}} + [(T_{\text{ambient}} - T_{\text{old}})/\tau]dt$
Fuel update	$m_{\text{new}} = \max(m_{\text{old}} - \text{burn_rate} \cdot dt, 0)$

20. Important code-name check

In the code shown in the conversation, the controller is written as **Plcontroller** and **compute_flow()**. Earlier project code used **PIController** and **compute()**. Python is case-sensitive, so the import/class/method names must match the actual controller.py file exactly.

Core mental model: Flight conditions change pressure and temperature; fuel burn changes ullage volume; these change the required gas state; the PI controller reacts to O_2 concentration by changing NEA flow; the inerting model updates the gas composition and repeats the cycle.